

A domain-driven phonological analysis on sandhi-voicing in Modern Burmese

Tianyi Ni (ni.386@osu.edu)

The Ohio State University

SEALS 31, University of Hawai'i at Mānoa

1 Introduction

- There are abundant phonological phenomena when forming compounds in Burmese
 - ◇ Reduction: allows major syllables to be reduced to minor syllables in *certain* circumstances.
 - * e.g., [Burgdorf \(2020\)](#)
 - ◇ Sandhi-Voicing: allows the onset of syllables turn to voiced from voiceless in *certain* circumstances.
 - * e.g., [Nishi \(1998\)](#); [Soe \(1999\)](#); [Vittrant \(2012\)](#); [Ni \(2022\)](#)
 - ◇ Weak vowel deletion: schwa sound [ə] will be deleted when a compound is formed in some *certain* phonotactic circumstances.
 - * e.g., [Lee \(2013\)](#)

TWO QUESTIONS

- Theoretical Question 1: Under what circumstances will the sandhi-voicing be triggered?
- Theoretical Question 2: How does the internal morphological structure of these compounds with sandhi-voicing look like? (i.e., some restrictions on the lexical categories of the morphemes?)

- Some findings, to be motivated
 - ◇ [Okell \(1969\)](#): The *aspirated* obstruents are not voiceable after an [ə] sound;
 - ◇ [Nishi \(1998\)](#): The preceding syllables/morphemes must end in *nasal* consonant;
 - ◇ [Soe \(1999\)](#): The following syllables/morphemes must be preceded by an *open* syllable;
 - ◇ [Vittrant \(2012\)](#): The lexical meaning of compounds must be *decomposable*.

Phonological rule SUMMARY

- (1) $\begin{bmatrix} -\text{VOICE} \\ \text{OBSTRUENT} \end{bmatrix} \rightarrow \begin{bmatrix} +\text{VOICE} \end{bmatrix} / \begin{bmatrix} \text{NASAL} \end{bmatrix} ___$
- (2) $\begin{bmatrix} -\text{VOICE} \\ \text{OBSTRUENT} \end{bmatrix} \rightarrow \begin{bmatrix} +\text{VOICE} \end{bmatrix} / \text{V} ___$
- (3) $\begin{bmatrix} -\text{VOICE} \\ +\text{ASPIRATE} \end{bmatrix} \rightarrow \begin{bmatrix} +\text{VOICE} \end{bmatrix} / [\text{ə}] ___$

One further QUESTION

- Empirical Question: How valid are these generalizations so far?

2 Internal morphological structure of compounds in Burmese

- Modern Burmese compounds are morphologically complex (see also [Burgdorf 2020](#))
 - ◇ VP compounds ← verb-verb
 - ◇ NP compounds ← NP-coordinate, NP-NP (noun-noun), NP-VP (noun-verb), VP-NP (verb-noun)

❶ VP compounds: There is always at least *one* verb in the compounds

- ◇ Observe that under the same phonological conditions, it is possible to get both **sandhi-absence** and **sandhi-voicing**

(4) VERB-VERB

- a. ne + t^ham → ne t^ham stay + sit → reside
- b. ti + s^hauʔ → ti s^hauʔ build + build → build
- c. ko + t^heʔ → ko d^zeʔ fry + cook → stir-fry

- ◇ The two consecutive verbs seem to form **serial verb construction**, for **both** cases
 - ✱ i.e., there is **no** morphological difference between two verb compounds where **sandhi-absence** and **sandhi-voicing** occur, respectively.

❷ NP compounds

- ◇ Observe that at least one noun is included

- ◇ It is possible to get both **sandhi-absence** and **sandhi-voicing** under the same morphological structure

(5) NOUN-NOUN

a.	sa + pe → sa pe	text + palm leaf → literature	N[[N text] [N palm leaf]]
b.	mwe + s ^h eiʔ → mwes ^h eiʔ	snake + poison → venom	N[modifier snake [N-head poison]]
c.	əmi + əp ^h a → mi ba	mother + father → parents	N[[N mother] [N father]]
d.	s ^h è + ək ^h àn → s ^h èg ^h àn	medicine + room → clinic	N[modifier medicine [N-head room]]

⇒ Similar to VERB-VERB in VP compounds, there is **no** morphological difference between NOUN-NOUN compounds where **sandhi-absence** and **sandhi-voicing** occur, respectively.

(6) VERB-NOUN

a.	pja + taiʔ → pja d aiʔ	show + building → museum	N[modifier show [N-head building]]
b.	gəzá + kwín → gəzá g wín	play + field → playground	N[modifier play [N-head field]]

⇒ VERB-NOUN NP compounds are very limited in verb behaving as a modifier, while the following nouns are always N-head.

- * i.e., **sandhi-absence** is **not** found in our data;
- * i.e., the onset of the following nouns are always voiced

(7) NOUN-VERB

a.	siʔ + pjan → siʔp j an	war + return → veteran	N[modifier war [N-head return(-er)]]
b.	sa + kù → sa k ù	text + copy → copy	N[modifier text [N-head copy]]
c.	p ^h òun + tɛ̀ → p ^h òun t ɛ̀	merit + be great → monk	N[N-head merit [modifier be great]]
d.	s ^h i + tɕ ^h ɛʔ → s ^h i ɕ ɛʔ	oil + cook → cooked/used oil	N[N-head oil [modifier cook]]
e.	mì + tɕ ^h ɪʔ → mì ɕ ɪʔ	fire + scratch → lighter	N[modifier fire [N-head scratch(-er)]]

⇒ There is **no** morphological difference between NOUN-VERB compounds where **sandhi-absence** and **sandhi-voicing** occur, respectively.

- * **Note:** Even though the left-headed case occurs far less than the right-headed case, the fact shown above still shows we know nothing how to distinguish one from the other from the compound morphology.

- **Back to Empirical Question**, the phonological rules proposed previously are not really accurate, because even if the phonotactic conditions are met, the sandhi-voicing is **not** guaranteed
- **Back to Thoretical Question 2**, we have analyze the internal morphological structures of both types of compounds in Burmese.
 - ◇ It seems that morphological structure cannot be seen as one of the cues, telling the speakers when to have the sandhi-voicing.

- ◇ The **exception** is for the VERB-NOUN compounds, the preceding verb is always the modifier, followed by the N-head with its onset voiced.

OBSERVATION

- **Observation 1:** It seems that in some compounds, the morpheme boundary is not approachable for the phonological processes
- **Observation 2:** Under certain circumstances, where all the phonotactic conditions are all met, as shown in (1)-(3), we can tell
 - ◇ Unlike what Vittrant (2012) proposed, some lexical meaning of compounds are decomposable, but sandhi-voicing still does not occur
 - ◇ On the other hand, some lexical meaning of compounds are not decomposable, and sandhi-voicing is not found, as Vittrant (2012) proposed

⇒ HYPOTHESIS: The seeming compounds, which do not show any sandhi-voicing, are essentially different from those with sandhi-voicing.

⇒ LINKING HYPOTHESES: These types of words have different DOMAINS, when they are stored in the speakers' lexicon

- ◇ PHRASES
- ◇ TYPICAL COMPOUNDS, hereafter COMPOUNDS
- ◇ LEXICALIZED WORDS, hereafter LEWD.

influenced by Archangeli & Pulleyblank (2022); Vittrant (2012)

3 Domain and emergence

- Emergent Phonology theory believes that a *morph* is a string of sounds associated with a meaning and/or grammatical function.
 - ◇ e.g., in Burmese, a mono-syllabic free or bounded morpheme, or a longer morpheme, or an unbreakable loanword can be seen as a morph.
- A morph should be the smallest unit that is acquired and stored for the *first* time.
- Two or more morphs with same morphological or phonological features form a larger data structure: *morph set* labelled by the shared features, such as lexical category, or voicing, etc.

◇ e.g., $\{\{t\epsilon\epsilon\}, \{dz\epsilon\}\}_{\text{TO.COOK}}, \{\text{sáun}, \text{záun}\}_{\text{TO.WATCH}}, \dots\}_{\text{VERB}}$

- Hypotheses Decomposition

- ◇ PHRASES consist of multiple morphs. Forming a new PHRASE is a syntactic process. Domain contains *two* words.
- ◇ COMPOUNDS also consist of multiple morphs, but forming a new COMPOUND is a morphological process. Domain contains *three* words.
- ◇ LEWD *per se* is a morph, which can be used to form new PHRASES and COMPOUNDS. Forming a new LEWD is a diachronic process. Domain contains only *one* word, itself.

★ Evidence:

(8) $\text{nay} + \text{pyi} + \text{taw} \rightarrow \text{naypyidaw}$ $\text{sun} + \text{land} + \text{royal} \rightarrow \text{royal palace}$

- Based on the phonological rules (1)-(3), we can generate the two phonotactic conditions, as shown in (9) and (10).

(9) **No unvoiced obstruent after open syllables.**

$*V \begin{bmatrix} \text{NONVOICE} \\ \text{OBSTRUENT} \end{bmatrix}$ With a focus on consonants, assign a violation to a word if an unvoiced obstruent is preceded by a long vowel.
 $\mathcal{F}$: consonant
 $\mathcal{D}$: word

(10) **No unvoiced obstruent after nasal coda.**

$* \begin{bmatrix} \text{CONS} \\ \text{NASAL} \end{bmatrix} \begin{bmatrix} \text{NONVOICE} \\ \text{OBSTRUENT} \end{bmatrix}$ With a focus on consonants, assign a violation to a word if an unvoiced obstruent is preceded by a nasal consonant.
 $\mathcal{F}$: consonant
 $\mathcal{D}$: word

- The word-initial voiceless consonant is voiced only if the voicing conditions are met, their corresponding counterparts with word-initial voiced consonant are the less frequent morphs.
 - ◇ e.g., the less frequent forms, such as $*\{\text{dəmín}\}_{\text{RICE}}, * \{dz\epsilon\}_{\text{TO.COOK}}$ will be prohibited, since the language learners are unlikely to hear them.
- Then, condition (11) is proposed, accordingly.

(11) **Burmese word-initial phonotactics**

*# [VOICE] With a focus on consonants, assign a violation to a word containing an
 $\mathcal{F}$: consonant initial voiced consonant.
 $\mathcal{D}$: word

- With all the three conditions in hand, we are able to assess compounds with [sandhi-voicing](#), as shown in (12):

(12) **Assessment for** [t^həmìn-dzɛɪ]_{CHEF-NOUN}
morph sets: {t^həmìn, dəmìn}_{RICE}; {tɛɪ, dzɛɪ}_{TO.COOK}

RICE-TO.COOK	*V	NONVOICE OBSTRUENT	*	CONS NASAL	NONVOICE OBSTRUENT	*# [VOICE]
☞ a. t ^h əmìn-dzɛɪ						*
b. dəmìn-dzɛɪ						**!
c. t ^h əmìn-tɛɪ					*!	
d. dəmìn-tɛɪ					*!	*

- For the LEWD that is not decomposable, without our hypothesis, we can get the assessment as shown in (13):

(13) **Tentative assessment for** [s^hóun-p^hya]_{TO.DECIDE-VERB}
morph sets: {s^hóun, zóun}_{TO.END-VERB}; {p^hya, bya}_{TO.CUT-VERB}

TO.DECIDE	*V	NONVOICE OBSTRUENT	*	CONS NASAL	NONVOICE OBSTRUENT	*# [VOICE]
a. s ^h óun-p ^h ya					*!	
b. zóun-p ^h ya					*	*!
☞ c. s ^h óun-bya						*
d. zóun-bya						**!

- This proves that these LEWDs are actually stored in a different way from COMPOUNDS in this model. The updated assessment table is given below:

(14) **Updated assessment for** [s^hóunp^hya]_{TO.DECIDE-VERB}
morph sets: {s^hóunp^hya, zóunp^hya}_{TO.DECIDE-VERB}

TO.DECIDE	*V	NONVOICE OBSTRUENT	*	CONS NASAL	NONVOICE OBSTRUENT	*# [VOICE]
☞ a. s ^h óunp ^h ya						
b. zóunp ^h ya						*!

(15) **Modern Burmese VERB morph set**

{ {s^hóunp^hya, zóunp^hya }_{TO.DECIDE}, {s^hóun, zóun }_{TO.END}, {p^hya, bya }_{TO.CUT}, {gəzà }_{TO.PLAY}, ... }_{VERB}

References

- Archangeli, D., & Pulleyblank, D. (2022). *Emergent Phonology*. Berlin: Language Science Press: Conceptual Foundations of Language Science.
- Burgdorf, D. C. (2020). Reduction in burmese compounds. *Journal of the Southeast Asian Linguistics Society*, 13.
- Lee, M. (2013). Phonology of noun subcompounding in burmese: Rules and their interactions. *Linguistic Science Research*, 66.
- Ni, T. (2022). Burmese sandhi-voicing: From the perspective of emergent phonology. *Coyote paper 24: Proceedings of the Arizona Linguistics Circle 14*.
- Nishi, Y. (1998). The Development of Voicing Rules in Standard Burmese. *Bulletin of the National Museum of Ethnology*, 23, 253-260.
- Okell, J. (1969). *A Reference Grammar of Colloquial Burmese (two volumes)*. London: Oxford University Press.
- Soe, M. (1999). *A Grammar of Burmese*. University of Oregon.
- Vittrant, A. (2012). How typology allows for a new analysis of the verb phrase in burmese. *LIDIL - Revue de linguistique et de didactique des langues, UGA Editions*, 46, 101-126.